

# TRIASHA SARKAR

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Machine Learning Engineer with experience in scientific ML, model evaluation, retrieval, time-series data, and ML systems. Aerospace background with experience in simulation, uncertainty, and engineering problems.

## EDUCATION

### MS, Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA | Aug 2026

CGPA: 3.7/4.0 | Entered PhD program; transitioned to MS and completed the degree in Aug 2026

### MSc, Aerospace Vehicle Design | Cranfield University, United Kingdom | Feb 2023

CGPA: 3.6/4.0 | First Class Honours | Thesis on geometry optimization under FAA constraints

### B.Tech, Aerospace Engineering | SRM Institute of Science and Technology, India | Jun 2021

CGPA: 3.8/4.0 | First Class Honours

## TECHNICAL SKILLS

ML and scientific ML: PyTorch, PyTorch Geometric, scikit-learn, graph neural networks, neural operators, surrogate modeling, uncertainty quantification, Transformers, RAG, BM25, dense retrieval, reranking

Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, PostgreSQL/pgvector, Prometheus, OpenTelemetry, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, CFD meshes, NetworkX, OSMnx, statistical inference, design of experiments

## PROFESSIONAL EXPERIENCE

### Graduate Research Assistant under Prof. Dimitri Mavris | Georgia Tech ASDL | May 2025 – Aug 2026

- Built and tested Python models for projects in simulation, airline safety, and sustainable aviation; shared reproducible analyses with faculty and sponsors.
- Reviewed assumptions and limits with aerospace experts and documented each workflow so other researchers could rerun it.

### Machine Learning Engineer | Rolls-Royce | Jul 2023 – Apr 2025

- Developed Python pipelines to flag unusual engine behavior, support maintenance planning, and combine sensor streams sampled at different rates.
- Mapped certification requirements to data and model checks, then reviewed results with lifecycle engineers before use.

### Data Science Intern | Rolls-Royce DataLabs | May 2021 – Jul 2021

- Cleaned aircraft-engine sensor data, built features, and compared prediction and anomaly models; summarized recurring failure patterns for engineers.

## TECHNICAL PROJECTS

### EdgeGenBench | Real-flight ML and on-device inference | Aug 2026 – Present

- Trained a flight-anomaly model on NASA DASHlink recordings and tested it on 17,780 approaches from held-out aircraft. Its 0.738 macro F1 missed the release target.
- Exported the model to ONNX and tested corrupted sensor inputs; prediction agreement stayed above 99.55% across the tested cases.

### IntegrityBench | Moderation evaluation and release controls | Aug 2026 – Present

Built controlled evaluation and release checks for changing moderation policies, keeping separate candidates and frozen benchmark results distinct from deployment claims.

### AeroSynth-Eval | Synthetic inspection-image evaluation | Aug 2026 – Present

Evaluated synthetic aircraft-inspection image data with reproducible splits; reported where results were strong, limited, or still pending validation.

### NewsLens | Recommendation and real-time search | Jul 2026 – Aug 2026

- Used chronological splits to keep future clicks out of training. The recommender scored 0.366 NDCG@10, but the uncertainty interval still included no improvement.
- Built a Go, Kafka, PostgreSQL, and FastAPI ingestion path for new articles. In a 500-event run, publish p99 was 44 ms and search freshness p95 was 79 ms.

### Equity Backtest | Walk-forward signal evaluation | Jul 2026 – Present

Built a cross-sectional equity-signal backtest with trading costs and safeguards against common look-ahead and selection biases.

### **Silent Failure Detection in Rocket-Motor Simulations | Georgia Tech ASDL | May 2026 – Jul 2026**

Traced silent failures in a 15,120-case rocket-motor study to an uncapped convergence loop; a leakage-safe classifier reached 96.8% accuracy on unseen geometries.

### **Surrogate Model Learning | Reliability under design shift | Feb 2026 – Present**

- Evaluated Gaussian-process and conventional surrogates on public airfoil, energy, and concrete data using grouped splits, multi-seed analysis, split-conformal intervals, and distance-to-training-domain guards.
- An airfoil Gaussian process reached  $R^2$  0.8145 on a physically grouped split; a 10-seed mean of  $0.8662 \pm 0.0680$  exposed split sensitivity. Nominal 90% intervals did not retain 90% coverage under design shift.

### **AIRFAANS | Geometry-aware CFD surrogates | Jan 2026 – Apr 2026**

- Compared an MLP, MeshGraphNet-style GNN, and point neural operator across three matched seeds and 200 AirfRANS interpolation meshes per model; MeshGraphNet had the lowest mean field and drag error.
- Used simulation-level splits and train-only normalization; Reynolds/AoA OOD, uncertainty, and active-learning experiments remain pending.

### **AeroRAG-X | Retrieval, evaluation, and ML systems | Oct 2025 – Present**

- Built a technical-knowledge system over 3,233 NASA report sections using hybrid retrieval, reranking, evidence checks, controlled citations, and containerized FastAPI services; retained negative experimental findings rather than claiming every change helped.

### **GREEN TEA and Project EAGLE | Sustainable aviation modeling | Sep 2025 – Apr 2026**

Built surrogate models to speed up sustainability calculations and connected fuel, demand, transport, and life-cycle inputs; GREEN TEA remains in sponsor use.

### **HERO | Source-aware retrieval for airline safety | Georgia Tech ASDL | May 2025 – Aug 2026**

Developed source-aware retrieval and evaluation methods for airline-safety analysis.

### **Supersonic Business Jets and Noise Mitigation | Cranfield University | Sep 2022 – Feb 2023**

Studied geometry, noise mitigation, high-speed aerodynamics, and design trade-offs.

### **Bird-Strike Impact Analysis | SRM Institute of Science and Technology | Jun 2020 – Jun 2021**

Used ANSYS to compare aircraft nose-cone material choices for bird-strike impact.

### **Ornithopter Aerodynamics | SRM Institute of Science and Technology | Sep 2019 – May 2020**

Used CFD to examine ornithopter wing area and load factor.

## **LEADERSHIP**

Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech | Aug 2025 – Aug 2026

Organizing Team | Aaruush, SRM Institute of Science and Technology | Jul 2019 – Sep 2019

- Helped organize a campus hackathon, gaining early exposure to collaborative, deadline-driven technical projects and the student developer community.

Hackathon Participant | SRM Institute of Science and Technology | Sep 2019

- Worked in a five-member team to plan a sustainable-city modeling project.